

Markov Model

- A Markov model is a stochastic model used to model randomly changing systems over time.

X_0 be the number of student entered in this classroom at time $t=t_0$

X_1 be the number of student entered in this classroom at time $t=t_0 + 1$

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X_n be the number of student entered in this classroom at time $t=t_n$

Let, $\{ X_n , n=0, 1, 2, \dots \}$ be a stochastic process that takes on a finite number of possible values.

Markov Model

Notational convention:

$$X_n = i$$

Where n is the time-stamp and i is the number of student (state).

Andrei Markov (1856-1922)

(A Russian mathematician)

<http://www-gap.dcs.st-and.ac.uk/~history/Mathematicians/Markov.html>

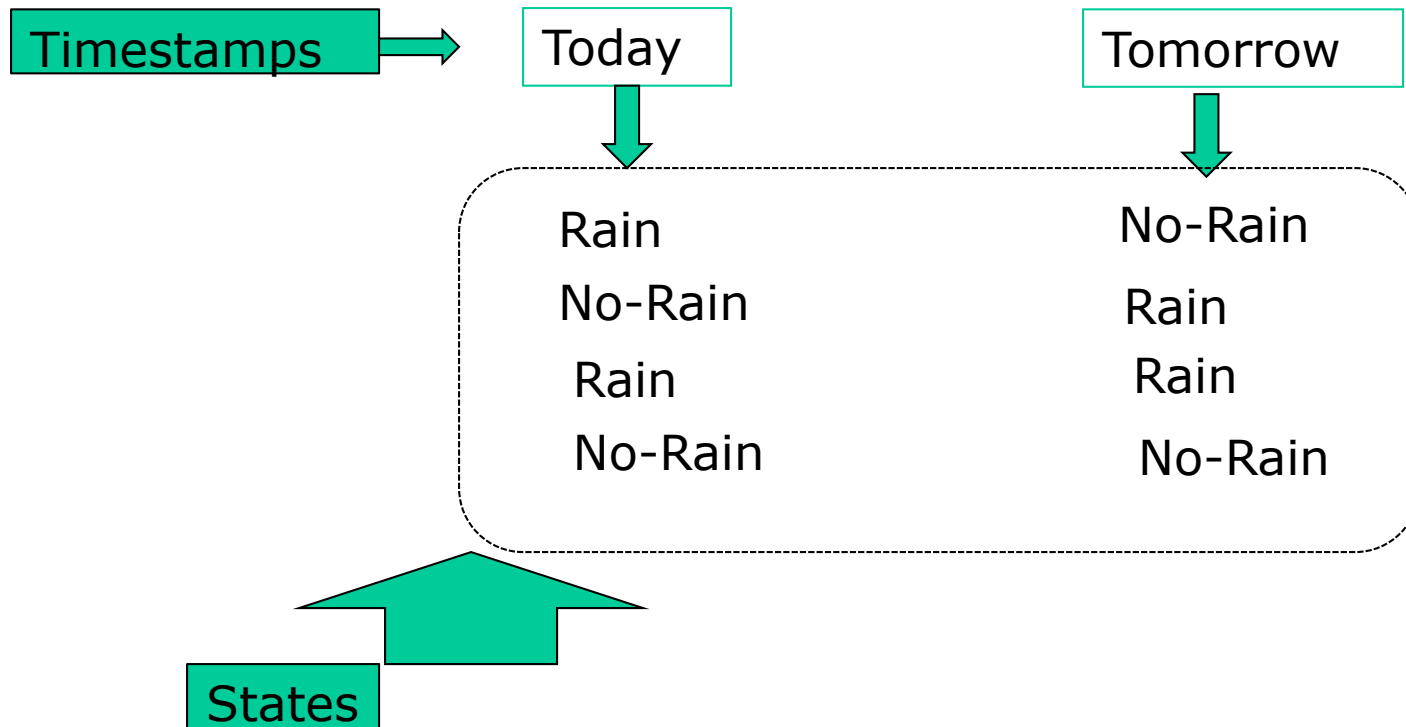
- Markov is particularly remembered for his study of Markov chains, sequences of random variables in which the future variable is determined by the present variable
- but is independent of the way in which the present state arose from its predecessors. This work launched the theory of stochastic processes



Terminologies of Markov Model

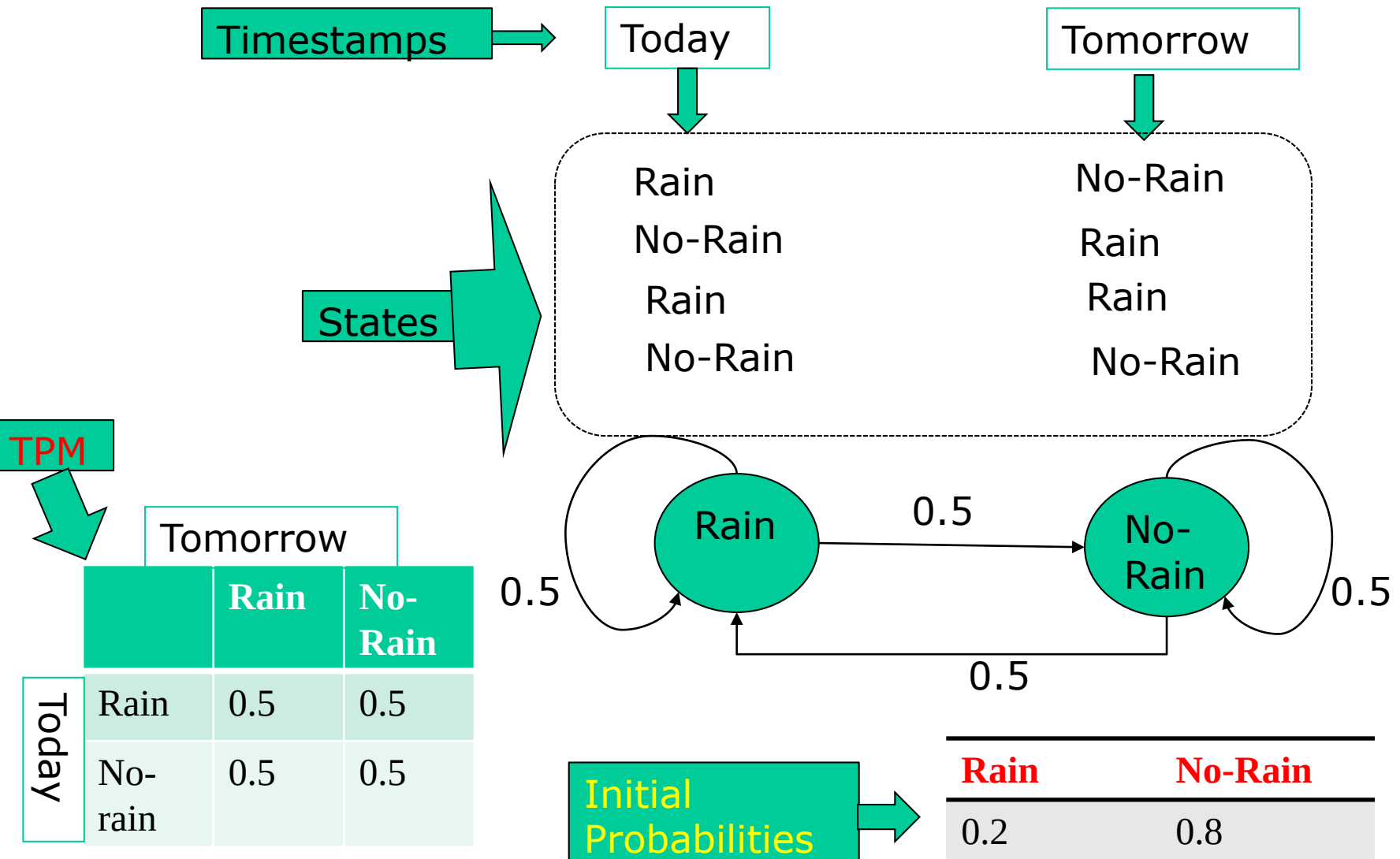
- States
- Transition Probabilities

States in MM



States and Transition Probabilities

Matrix (TPM) in MM



Markov Models

- Joint Distribution

$$\begin{aligned} p(\mathbf{X}) &= p(X_1, X_2, \dots, X_n) \\ &= p(X_1)p(X_2|X_1)p(X_3|X_2, X_1) \dots p(X_n|X_{n-1}, \dots, X_1) \\ &= \prod_{i=1}^n p(X_i|X_{i-1}, \dots, X_1) \end{aligned}$$

Chain rule

- Markov Assumption (m^{th} order)

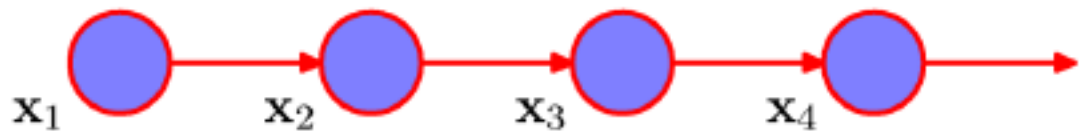
$$p(\mathbf{X}) = \prod_{i=1}^n p(X_i|X_{i-1}, \dots, X_{i-m})$$

Current observation only depends on past m observations

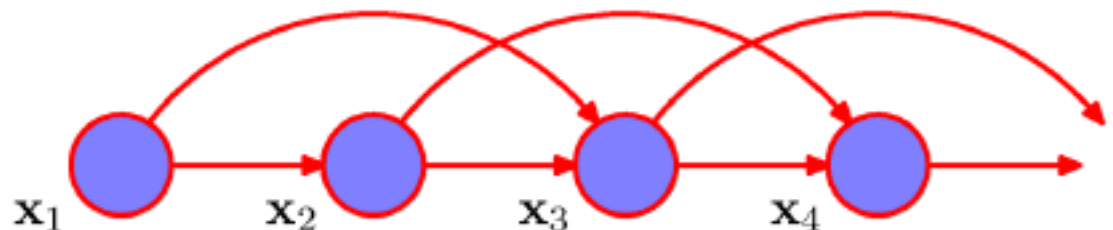
Markov Models

- Markov Assumption

1st order
$$p(\mathbf{X}) = \prod_{i=1}^n p(X_i | X_{i-1})$$



2nd order
$$p(\mathbf{X}) = \prod_{i=1}^n p(X_i | X_{i-1}, X_{i-2})$$



Markov Models

- Markov Assumption

$$\text{1}^{\text{st}} \text{ order} \quad p(\mathbf{X}) = \prod_{i=1}^n p(X_i | X_{i-1})$$

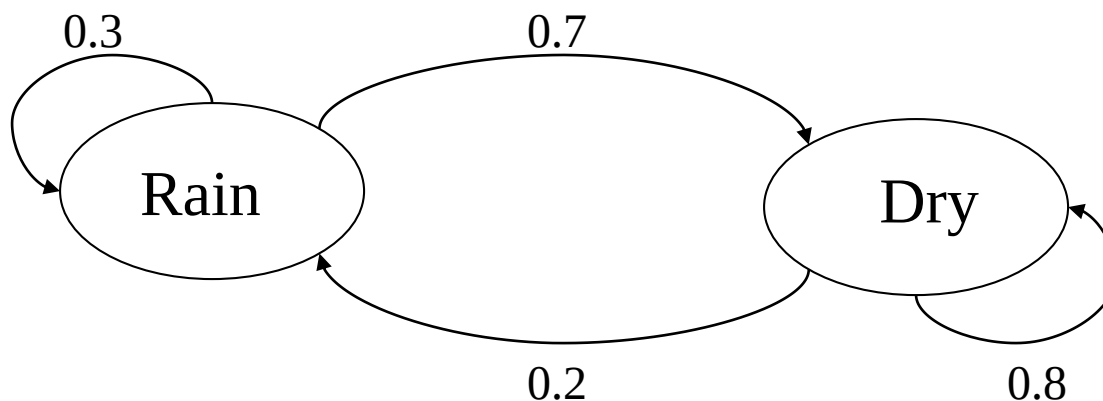
$$\text{m}^{\text{th}} \text{ order} \quad p(\mathbf{X}) = \prod_{i=1}^n p(X_i | X_{i-1}, \dots, X_{i-m})$$

$$\text{n-1}^{\text{th}} \text{ order} \quad p(\mathbf{X}) = \prod_{i=1}^n p(X_i | X_{i-1}, \dots, X_1)$$

≡ no assumptions – complete (but directed) graph

Homogeneous/stationary Markov model (probabilities don't depend on n)

Example of Markov Model



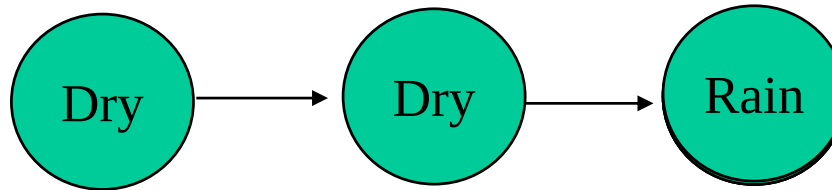
- Two states : ‘Rain’ and ‘Dry’.
- Transition probabilities:

$P(\text{‘Rain’}|\text{‘Rain’})=0.3$, $P(\text{‘Dry’}|\text{‘Rain’})=0.7$,

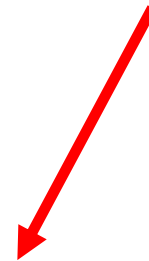
$P(\text{‘Rain’}|\text{‘Dry’})=0.2$, $P(\text{‘Dry’}|\text{‘Dry’})=0.8$

- Initial probabilities: say $P(\text{‘Rain’})=0.4$, $P(\text{‘Dry’})=0.6$.

Calculation of sequence probability



$$\begin{aligned} &P(S_1 = \text{Dry}, S_2 = \text{Dry}, S_3 = \text{Rain}) \\ &= \underline{P(S_1 = \text{Dry})P(S_2 = \text{Dry} \mid S_1 = \text{Dry})P(S_3 = \text{Rain} \mid S_2 = \text{Dry})} \\ &= 0.6 \times 0.8 \times 0.2 \end{aligned}$$



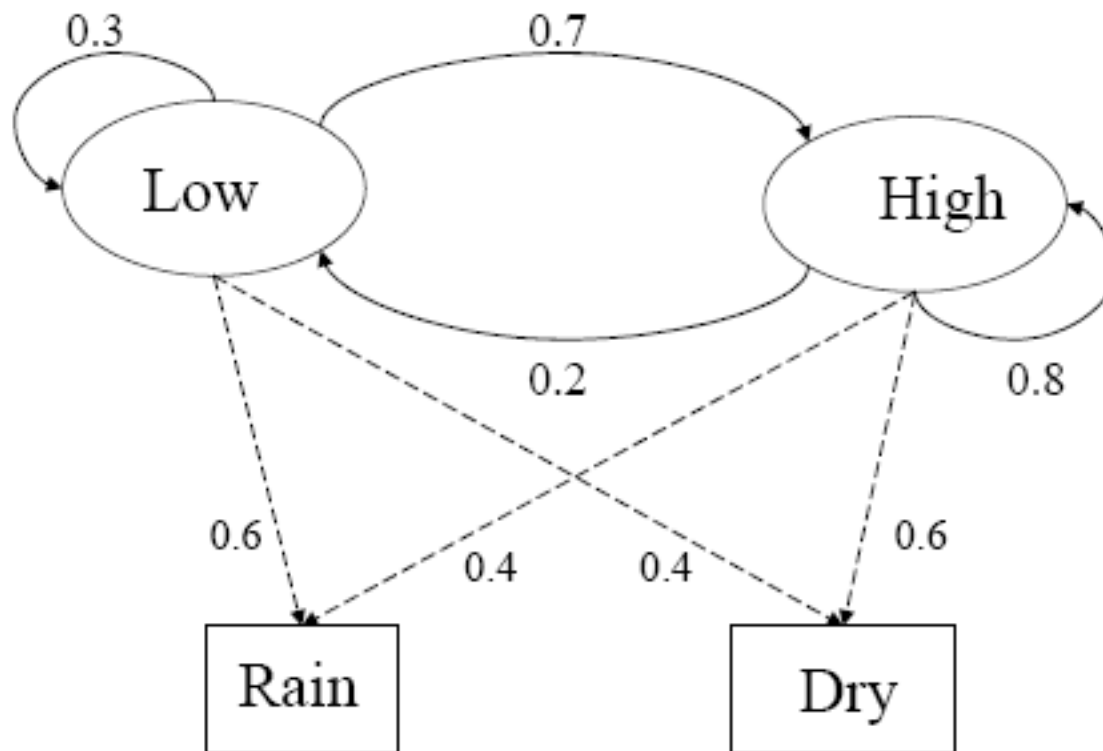
Markov Model is just a Bayesian Network;

In this network, $P(S_1, S_2, S_3) = P(S_1)P(S_2|S_1)P(S_3|S_2)$

Hidden Markov Models

- Based on Markov Models
- Differences include
 - State becomes “Hidden”
 - The state information is not available, instead of, there are some “Observations” which are correlated with “Hidden” State

Hidden Markov Model



Two states : 'Low' and 'High' atmospheric pressure.
Two observations : 'Rain' and 'Dry'.
Initial probabilities: $P(\text{'Low'})=0.4$, $P(\text{'High'})=0.6$.

Example 1

Transition probability matrix:

	Low	High
Low	0.3	0.7
High	0.2	0.8

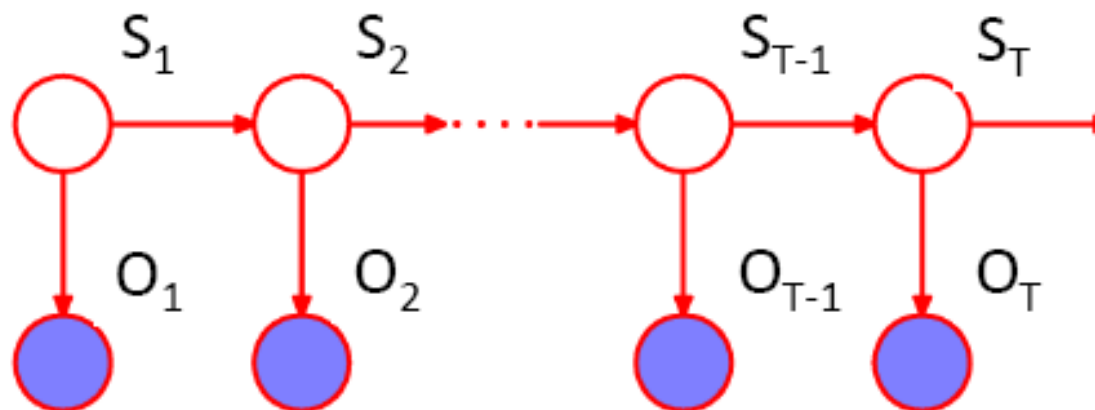
Emission probability matrix:

	Rain	Dry
Low	0.6	0.4
High	0.4	0.3

In Markov Model,
Emission Prof. is not used

Hidden Markov Models

- Distributions that characterize sequential data with few parameters but are not limited by strong Markov assumptions.



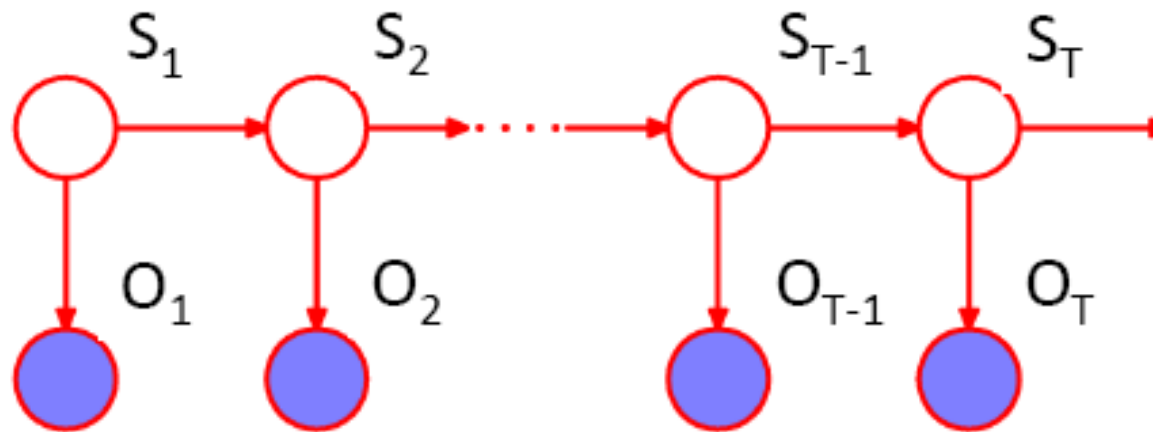
Observation space

$$O_t \in \{y_1, y_2, \dots, y_K\}$$

Hidden states

$$S_t \in \{1, \dots, I\}$$

Hidden Markov Models



$$p(S_1, \dots, S_T, O_1, \dots, O_T) = \prod_{t=1}^T p(O_t | S_t) \prod_{t=1}^T p(S_t | S_{t-1})$$

1st order Markov assumption on hidden states $\{S_t\}$ $t = 1, \dots, T$
(can be extended to higher order).

Hidden Markov Models

- Parameters – stationary/homogeneous markov model (independent of time t)

Initial probabilities

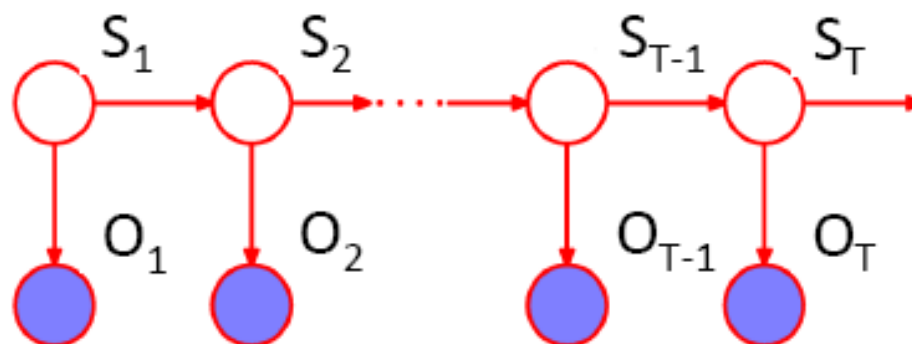
$$p(S_1 = i) = \pi_i$$

Transition probabilities

$$p(S_t = j | S_{t-1} = i) = p_{ij}$$

Emission probabilities

$$p(O_t = y | S_t = i) = q_i^y$$



$$p(\{S_t\}_{t=1}^T, \{O_t\}_{t=1}^T) = p(S_1) \prod_{t=2}^T p(S_t | S_{t-1}) \prod_{t=1}^T p(O_t | S_t)$$

Hidden Markov Models: Example

An experience in a casino

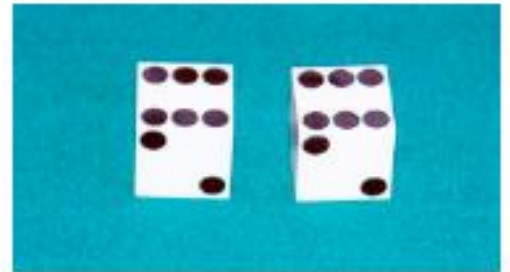
Game:

1. You bet \$1
2. You roll (always with a fair die)
3. Casino player rolls (sometimes with fair die, sometimes with loaded die)
4. Highest number wins \$2

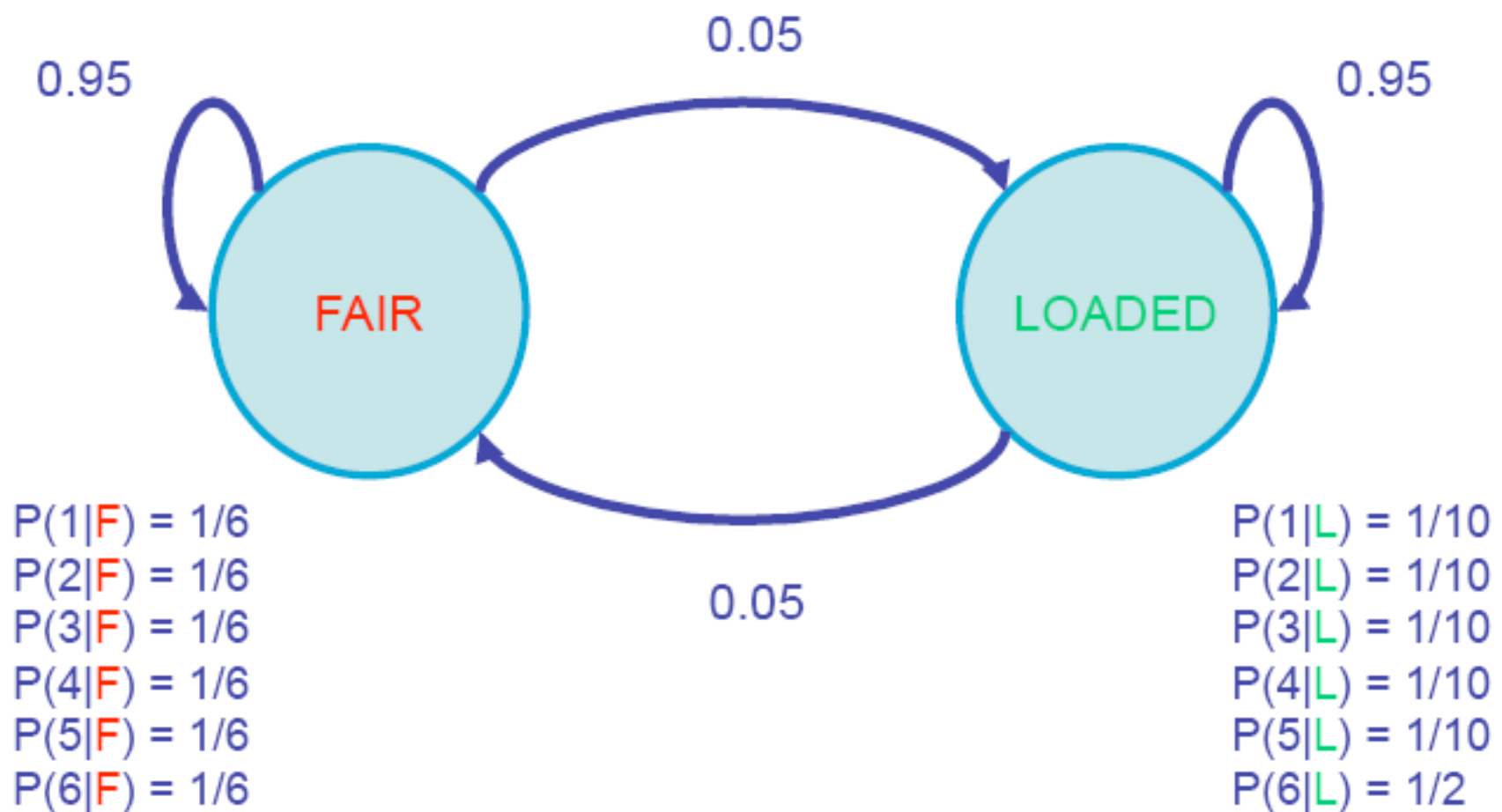
Here is **his** sequence of die rolls:

1245526462146146136136661664661636
616366163616515615115146123562344

Which die is being used in each play?



The Dishonest Casino Model



HMM Problems

GIVEN: A sequence of rolls by the casino player

1245526462146146136136661664661636616366163616515615115146123562344

QUESTION

- How likely is this sequence, given our model of how the casino works?
 - This is the **EVALUATION** problem in HMMs
- What portion of the sequence was generated with the fair die, and what portion with the loaded die?
 - This is the **DECODING** question in HMMs
- How “loaded” is the loaded die? How “fair” is the fair die? How often does the casino player change from fair to loaded, and back?
 - This is the **LEARNING** question in HMMs

Three main problems in HMMs

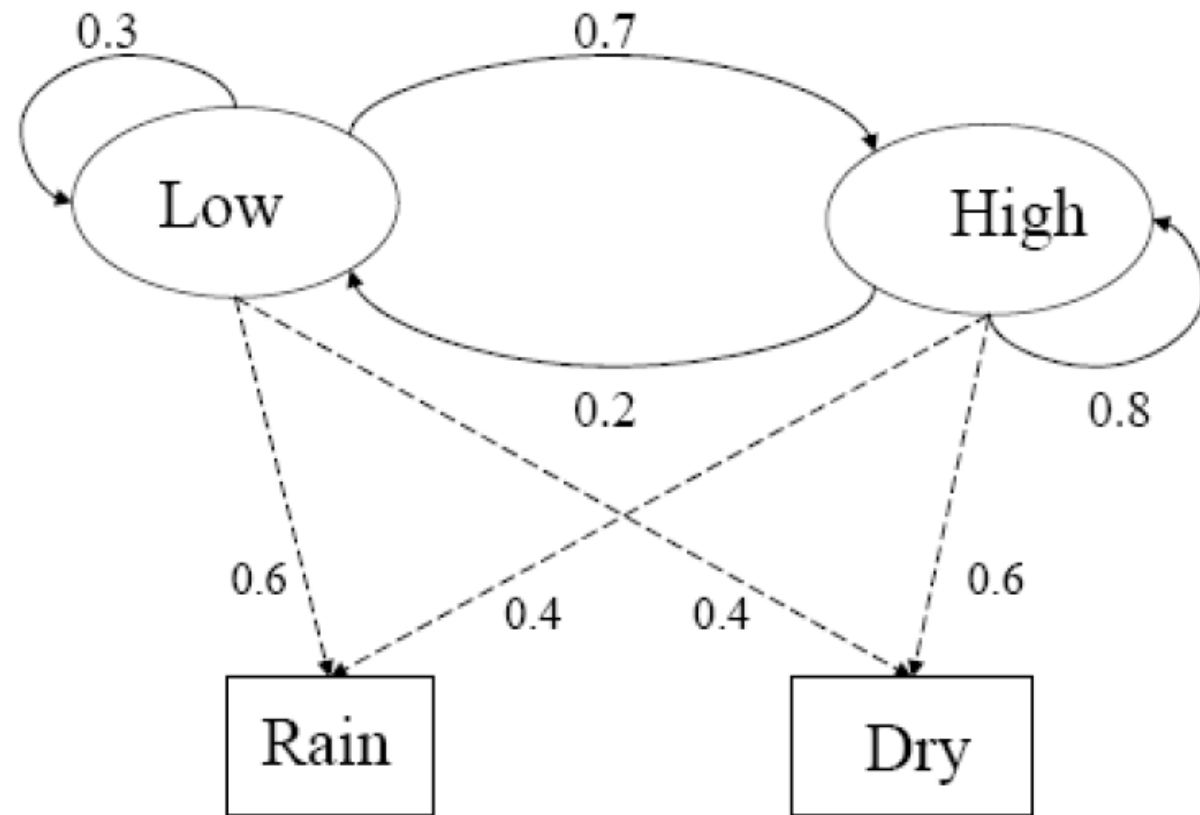
- **Evaluation** – Given HMM parameters & observation seqn $\{O_t\}_{t=1}^T$
find $p(\{O_t\}_{t=1}^T)$ prob of observed sequence
- **Decoding** – Given HMM parameters & observation seqn $\{O_t\}_{t=1}^T$
find $\arg \max_{s_1, \dots, s_T} p(\{S_t\}_{t=1}^T | \{O_t\}_{t=1}^T)$ most probable
sequence of hidden states
- **Learning** – Given HMM with unknown parameters and $\{O_t\}_{t=1}^T$
observation sequence
find $\arg \max_{\theta} p(\{O_t\}_{t=1}^T | \theta)$ parameters that maximize
likelihood of observed data

HMM Algorithms

- **Evaluation** – What is the probability of the observed sequence? **Forward Algorithm**
- **Decoding** – What is the probability that the third roll was loaded given the observed sequence? **Forward-Backward Algorithm**
 - What is the most likely die sequence given the observed sequence? **Viterbi Algorithm**
- **Learning** – Under what parameterization is the observed sequence most probable? **Baum-Welch Algorithm (EM)**

Hidden Markov Model

Example 1



Transition probability matrix:

	Low	High
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High	0.2	0.8

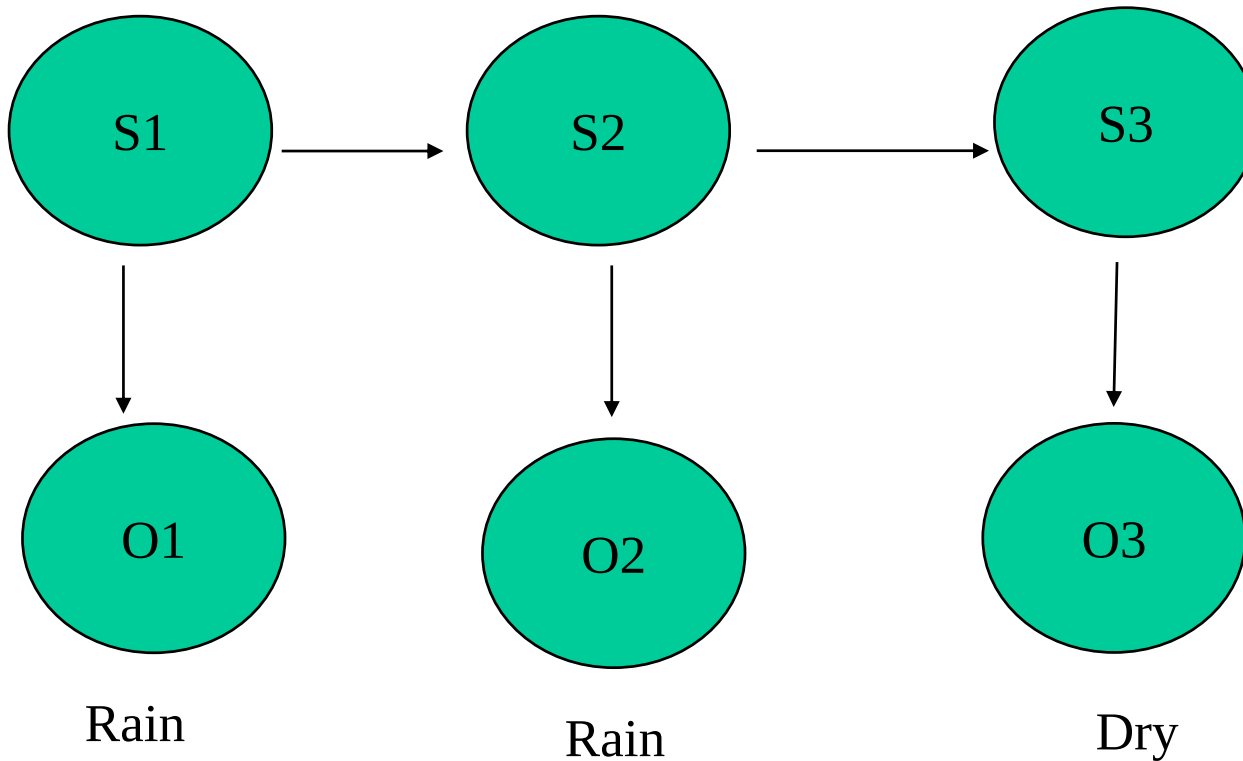
Emission probability matrix:

	Rain	Dry
Low	0.6	0.4
High	0.4	0.6

Two states : 'Low' and 'High' atmospheric pressure.
Two observations : 'Rain' and 'Dry'.
Initial probabilities: $P(\text{'Low'})=0.4$, $P(\text{'High'})=0.6$.

In Markov Model,
Emission Prof. is not used

Decoding Problem 2



O1,O2,O3 are known, what is the most probable sequence S1,S2,S3

For example: {High, High, Low}, {Low, High, Low}, {Low, High, High}.....

$$\arg \max_{S1, S2, S3} P(S1, S2, S3 | O1, O2, O3)$$

$$= \arg \max_{S1, S2, S3} P(S1, S2, S3, O1, O2, O3)$$

$$= \arg \max_k \max_{S1, S2} P(S3 = k, S1, S2, O1, O2, O3)$$

V_3^k

Probability of most likely sequence of states ending at states $S3=k$

$$V_3^k$$

$$= \max_{S1, S2} P(S3 = k, S2, S1, O1, O2, O3)$$

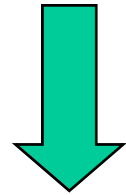
$$= \max_i \max_{S1} P(S3 = k, S2 = i, S1, O1, O2, O3)$$

$$= \max_i \max_{S1} P(S2 = i, S1, O1, O2) P(S3 = k, O3 | S2 = i, S1, O1, O2)$$

$$= \max_i \max_{S1} P(S2 = i, S1, O1, O2) P(O3 | S3 = k, S2 = i, S1, O1, O2) P(S3 = k | S2 = i, S1, O1, O2)$$



D-Separation



$$P(O3 | S3 = k)$$

$$P(S3 = k | S2 = i)$$

$$V_3^k = \max_i V_2^i P(O3 | S3 = k) P(S3 = k | S2 = i)$$

$$= P(O3 | S3 = k) \max_i P(S3 = k | S2 = i) V_2^i$$

Viterbi algorithm

$$\begin{aligned} V_3^k &= \max_i V_2^i P(O3 | S3 = k) P(S3 = k | S2 = i) \\ &= P(O3 | S3 = k) \max_i P(S3 = k | S2 = i) V_2^i \end{aligned}$$

$$V_2^k = P(O2 | S2 = k) \max_i P(S2 = k | S1 = i) V_1^i$$

Viterbi algorithm

Can compute V_t^k for all k, t using dynamic programming:

- Initialize: $V_1^k = p(O_1|S_1=k)p(S_1 = k)$ for all k

- Iterate: for $t = 2, \dots, T$

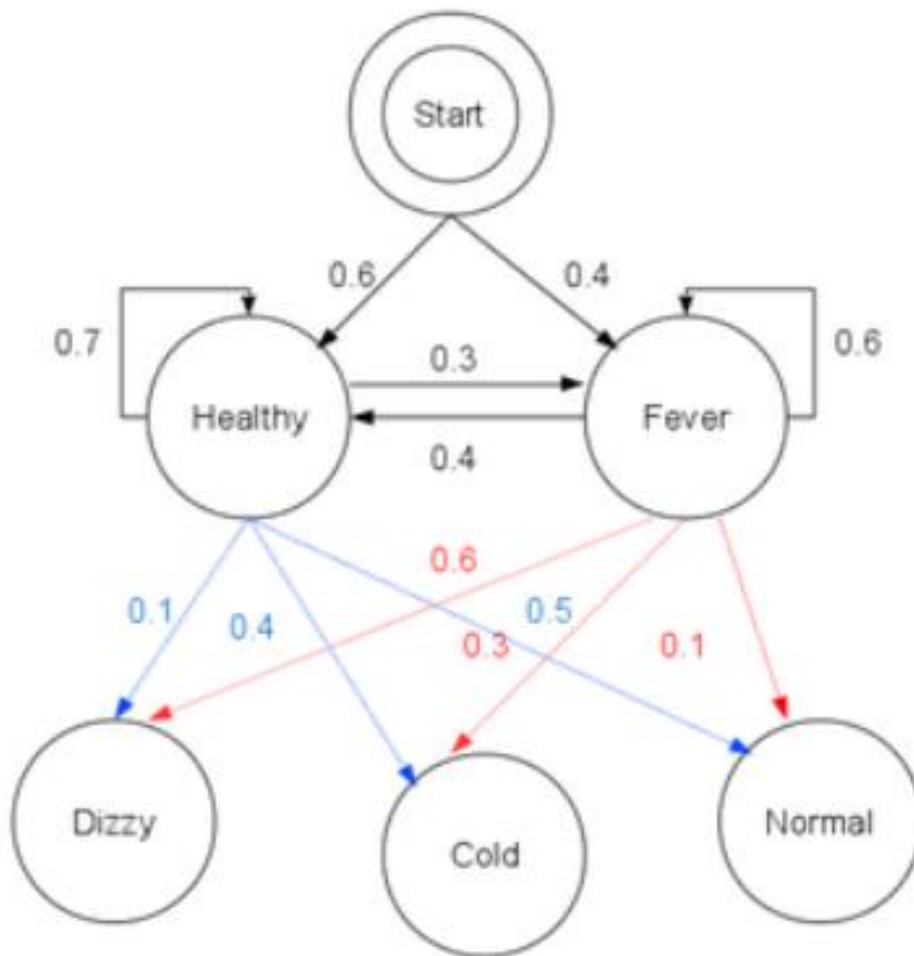
$$V_t^k = p(O_t|S_t = k) \max_i p(S_t = k|S_{t-1} = i) V_{t-1}^i \quad \text{for all } k$$

- Termination: $\max_{\{S_t\}_{t=1}^T} p(\{S_t\}_{t=1}^T, \{O_t\}_{t=1}^T) = \max_k V_T^k$

Traceback: $S_T^* = \arg \max_k V_T^k$

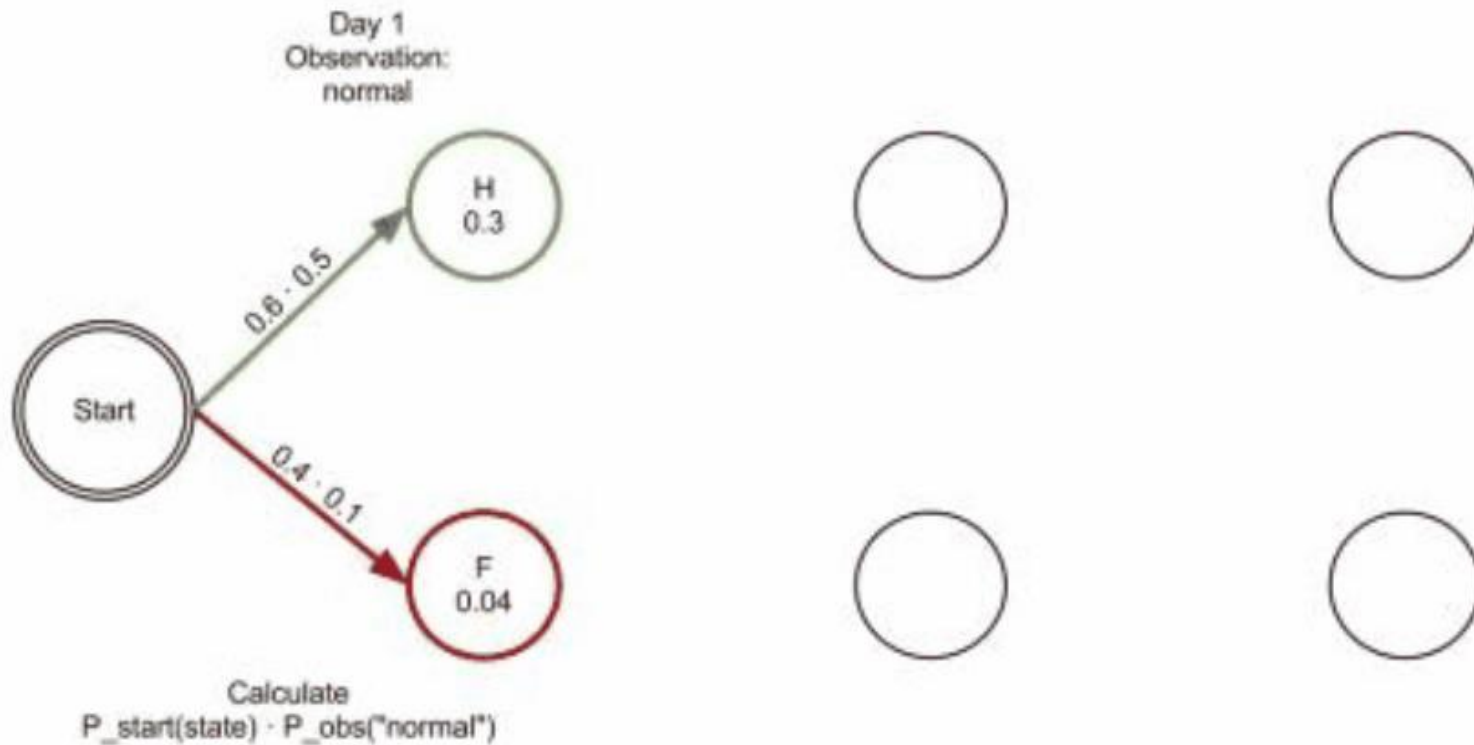
$$S_{t-1}^* = \arg \max_i p(S_t^*|S_{t-1} = i) V_{t-1}^i$$

HMM

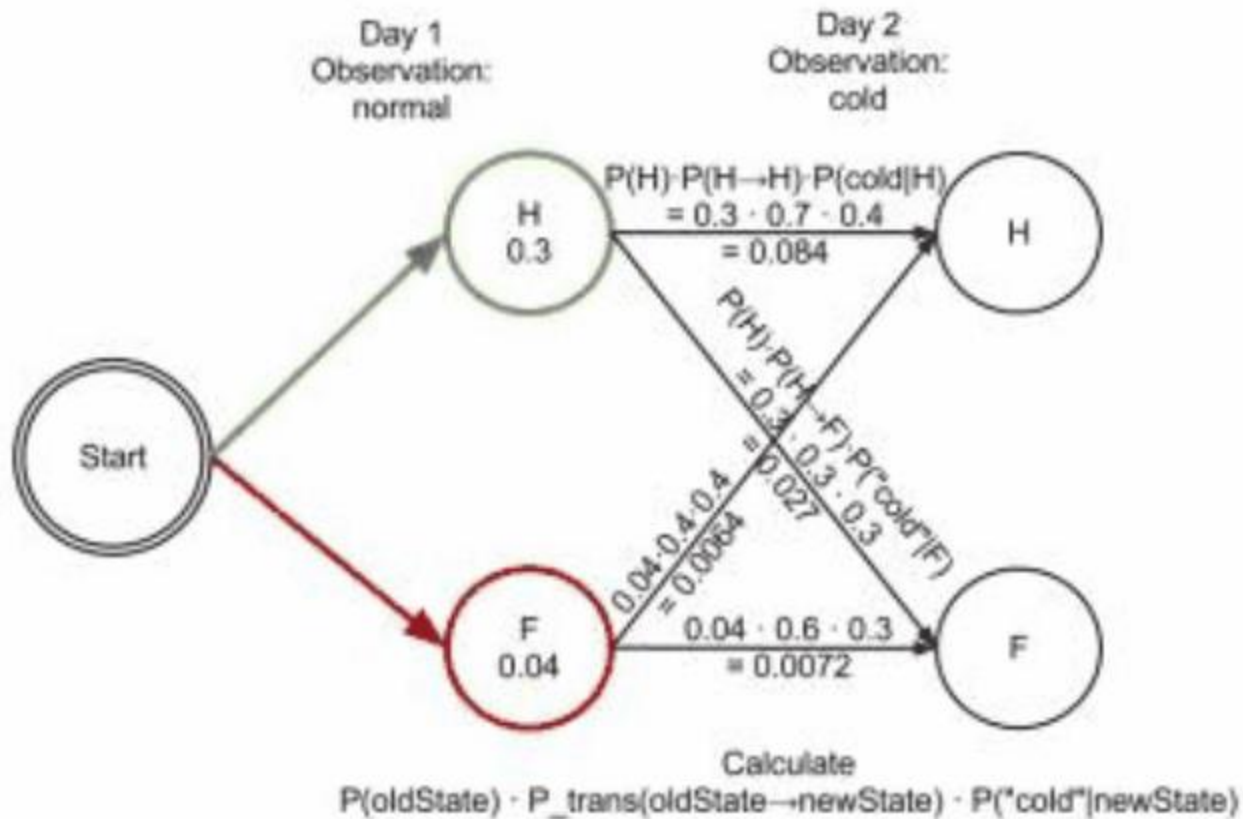


What is the most likely sequence of health status for the observation sequence **[Normal, Cold, Dizzy]**?

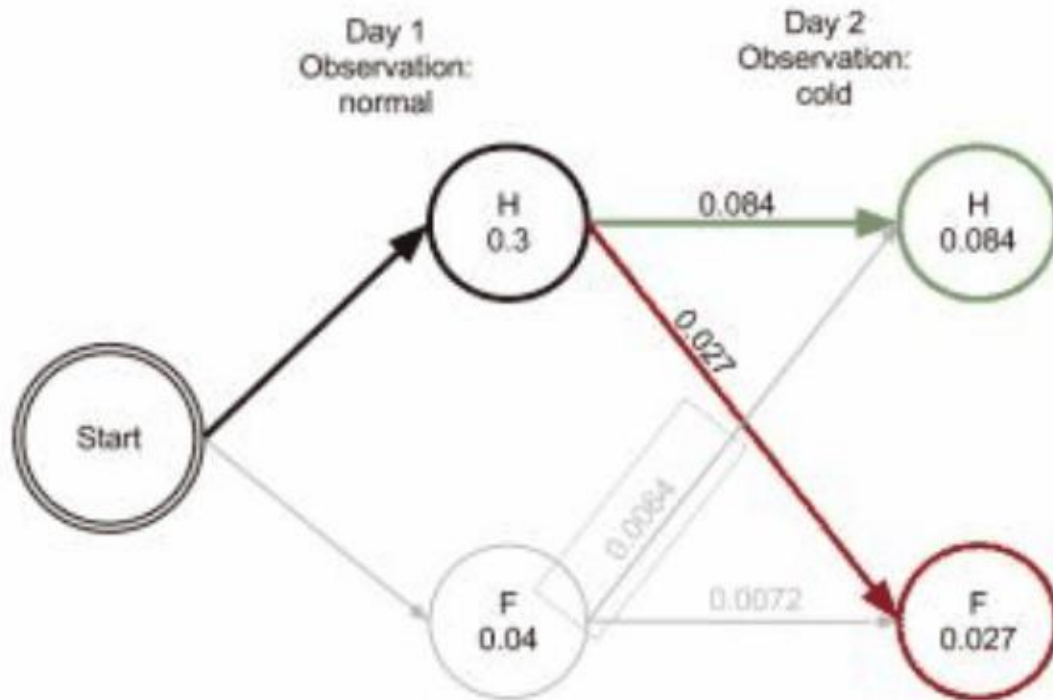
Viterbi Algorithm



Viterbi Algorithm

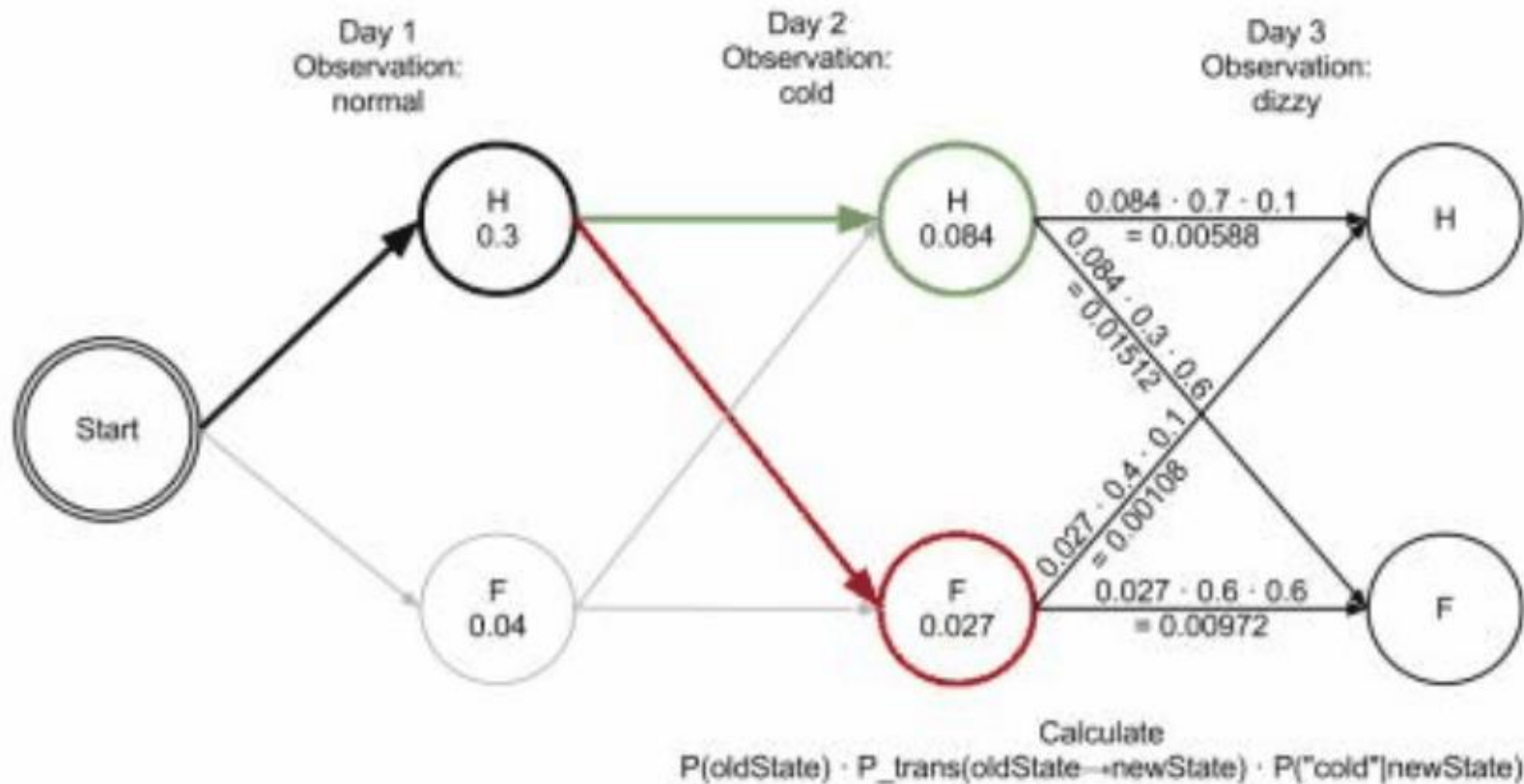


Viterbi Algorithm

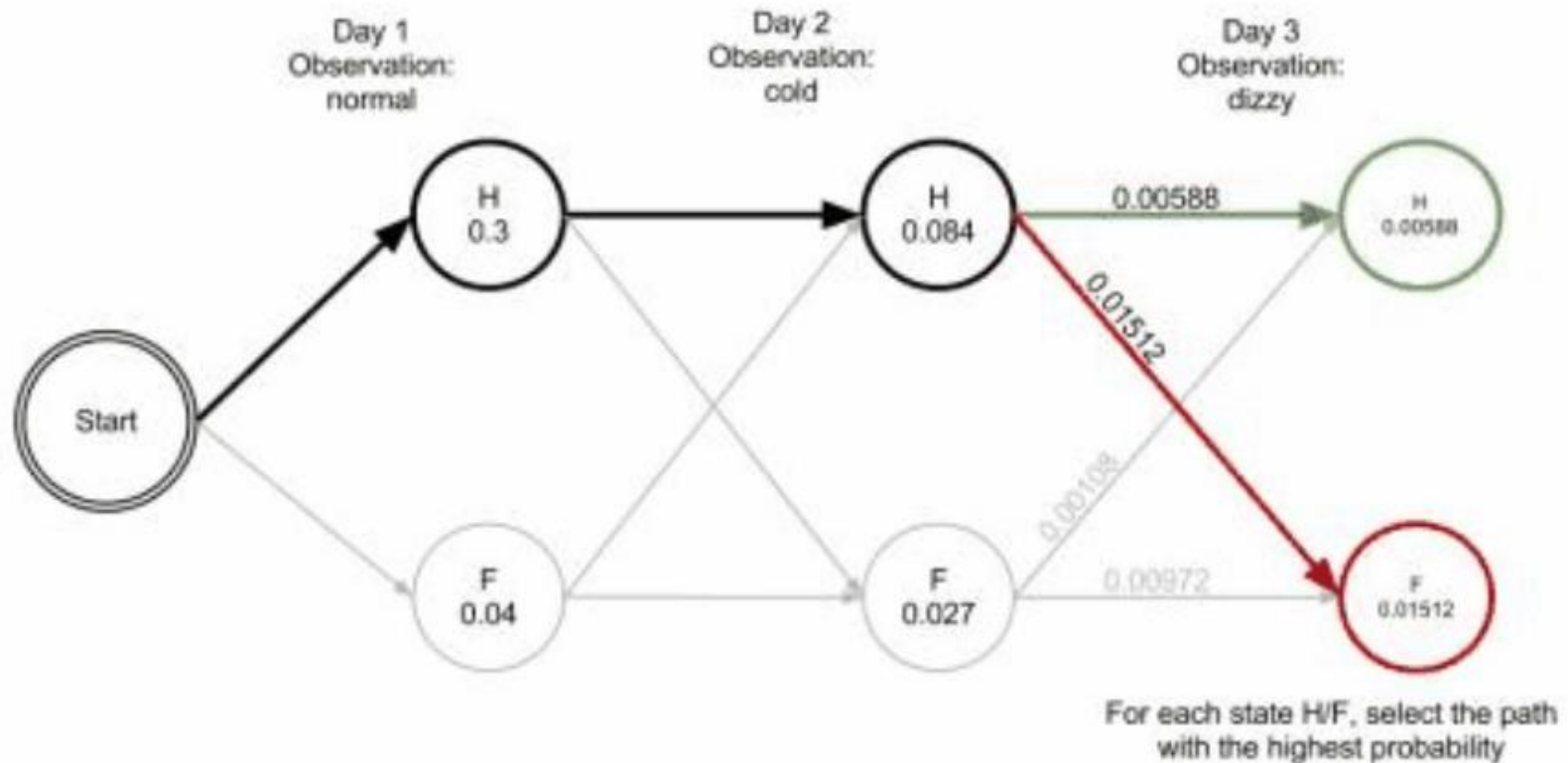


For each state H/F, select the path with the highest probability

Viterbi Algorithm



Viterbi Algorithm



The most likely sequence of health status is **[Health, Healthy, Fever]** for the observation sequence **[Normal, Cold, Dizzy]**.

Example Dataset

States: H, L

Observations: R, D

Timestamp →	1	2	3	4	5	6	7
State sequence 1	H	L	L	H	H	L	H
Obs sequence 1	R	R	D	D	R	D	R
State sequence 2	L	L	H	H	H	L	H
Obs sequence 2	D	R	D	D	R	R	D
State sequence 3	L	H	H	H	H	L	H
Obs sequence 3	R	D	D	D	R	D	R
State sequence 4	L	L	H	H	L	H	L
Obs sequence 4	R	R	R	D	D	R	D
State sequence 5	H	H	H	L	L	H	L
Obs sequence 5	D	R	D	D	D	D	R